

Answer **all** questions.

- 1** The Sunda slow loris (*Nycticebus coucang*) is a primate native to forests of Indonesia, western Malaysia, southern Thailand and Singapore. Listed as “Vulnerable” on the International Union for Conservation of Nature (IUCN) Red List, its population is fast declining due to extensive habitat loss and illegal poaching for pet trade.

In order to understand more about the species before it is extinct, molecular research work is conducted by a wildlife conservation research facility. A gene of interest from the species and a bacterial plasmid vector are cleaved using the same restriction enzyme and ligated before being cloned in a bacterial host model organism for further analysis. The restriction enzyme used in the experiment generates sticky ends while the bacterial plasmid used contains marker genes.

- (a) Explain how restriction enzymes cut DNA. [1]

- (b) Explain what is meant by *marker gene*. [1]

- (c) Suggest why it is possible to obtain two different types of recombinant plasmids in the experiment. [2]

The Sunda slow loris is protected by law under the Convention on International Trade in Endangered Species of Wild Fauna and Flora (CITES) signed by many countries. It is an offence to be in possession or to trade in any illegally imported or acquired CITES species.

Before the Sunda slow loris is sold as pet, its teeth are often removed to prevent injury to potential buyers. However, this frequently results in infection and death of the animal. In a recent search of the home of an alleged wildlife trader, no live animal was recovered except three removed teeth of suspected slow loris species.

DNA from dental pulp extracted from these teeth were analysed based on several Variable Number Tandem Repeat (VNTR) loci. Fig. 1.1 shows the genetic fingerprints of the three teeth found, labelled as Teeth 1, Teeth 2 and Teeth 3, and teeth known to be taken from a Sunda slow loris.

DNA ladder	Teeth from Sunda slow loris	Teeth 1	Teeth 2	Teeth 3
6 kb	_____		_____	
5 kb	_____			
4 kb				_____
3 kb	_____	_____	_____	_____
2 kb	_____		_____	
1 kb		_____		_____

Fig. 1.1

(d) (i) Briefly explain the basis of genetic fingerprinting.

[2]

- (ii) With reference to Fig. 1.1, which tooth is most likely removed from the Sunda slow loris? Explain your answer. [2]

- (iii) Rescued captive Sunda slow lorises that have their teeth removed could not be reintroduced into the wild, as they are unlikely to survive. Attempts at breeding Sunda slow lorises in captivity in zoos were made. However, animals continuously bred in captivity might experience inbreeding depression.

Using your biological knowledge on restriction fragment length polymorphism, explain how the banding patterns of a genetic fingerprint can show whether or not an animal is inbred. [3]

[Total: 11]

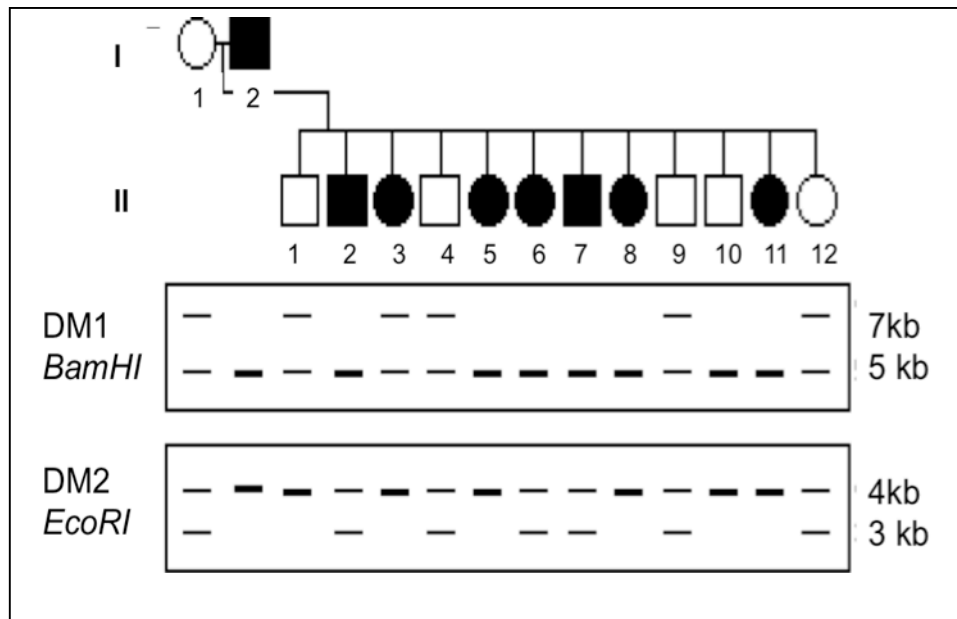
- 2 Cystic fibrosis (CF) is one of the most common genetic disorders in the United States. It is estimated that 30,000 Americans are living with CF and approximately 1,000 new cases are diagnosed every year. CF is caused by mutations in a gene located on chromosome 7.

(a) Explain how CF is inherited.

[2]

In the early 1980s, attempts were made to map the CF gene on chromosome 7 using DNA markers.

The relationship between the CF gene and two DNA markers DM1 and DM2 were studied in a family with many affected children. DNA were extracted from these individuals and cut with two different restriction enzymes, *Bam*HI and *Eco*RI, separately. The products of digestion were separated using gel electrophoresis and the RFLP banding patterns is shown in Fig. 2.1 below.



(Note: *Bam*HI was used on DM1 and *Eco*RI on DM2)

Fig. 2.1

- (b) (i)** Using the information provided in Fig. 2.1, deduce which marker is more suitable in detecting the presence of CF. [3]

- (ii)** Using your understanding of linked genes, provide an explanation for your answer to **(b)(i)**. [2]

- (c) An experiment using a non-viral vector to deliver the cystic fibrosis transmembrane conductance regulator (CFTR) gene was carried out on five individuals with CF. Three doses of plasmids bearing the CFTR gene were administered to the patients at four-week intervals. After each dose, tests were carried out on the patients to detect the following: plasmid-derived CFTR DNA, plasmid-derived CFTR mRNA, plasmid-derived CFTR protein and CFTR protein function. The results are shown in Table 2.2, with positive samples marked '+’.

Table 2.2

Patient	Dose 1				Dose 2				Dose 3			
	DNA	mRNA	protein	protein function	DNA	mRNA	protein	protein function	DNA	mRNA	protein	protein function
1	+	+	+	+			+		+			
2									+	+	+	+
3												
4	+	+	+		+	+	+					
5	+	+			+	+			+	+		

DNA : detection of plasmid-derived CFTR DNA
 mRNA : detection of plasmid-derived mRNA
 protein : detection of plasmid-derived CFTR protein
 protein function : detection of CFTR protein function

For (c) (i)-(iv), use the data from Table 2.2.

- (i) Calculate the percentage success rate of CFTR gene delivery by liposomes for all 15 doses. [1]

- (ii) Calculate the percentage success rate of transcription of plasmid-derived CFTR DNA for cases where gene delivery was successful. [1]

- (iii) From your answer in (c)(i) and (c)(ii), determine whether gene delivery or transcription of the delivered gene was the more significant obstacle to successful gene therapy. [1]

- (iv) Suggest an alternative delivery method that can help address the obstacle discussed in (c)(iii). [1]

- (d) Currently, gene therapy involves the introduction of the normal allele into somatic cells of individuals suffering from CF.

- (i) Explain why there seems to be fewer objections to this method compared to one where gene therapy is applied to gametic cells. [2]

- (ii) Explain why gene therapy provides a possible cure for CF but not stem cell therapy. [2]

[Total: 15]

- 3 A number of diseases, such as dengue fever, are spread by mosquitoes. The incidence of this disease has increased dramatically in recent years and this has been linked with the spread of the mosquito, *Aedes aegypti*.

In an attempt to reduce the numbers of *A. aegypti*, genetically modified (GM) male mosquitoes were produced. One of the genes added to these mosquitoes, **when switched on**, results in the production of a protein that is toxic to mosquitoes.

In 2010, in the Cayman Islands and in Malaysia, GM male mosquitoes were released into the wild to mate with females. All the resulting offspring died in the larval stage.

In Malaysia, both GM male and non-GM male mosquitoes were released in order to compare their dispersal and life span in the wild. The GM mosquitoes could be identified because they also carried a gene for green fluorescent protein (GFP).

- (a) Explain why, in many examples of gene technology, fluorescent markers are used in preference to antibiotic resistance genes. [1]

- (b) In addition to the gene for GFP, the DNA that has been added to the GM mosquitoes consists of
- a promoter
 - a gene coding for a toxic protein, tTA
 - a binding site for tTA.

When a GM mosquito larva hatches from an egg, the promoter induces the production of only a small amount of tTA, so that the larva does not die immediately. In a process of positive feedback, the tTA produced binds to the DNA as shown in Fig. 3.1. This increases the expression of the gene until the increased concentration of tTA kills the larva.

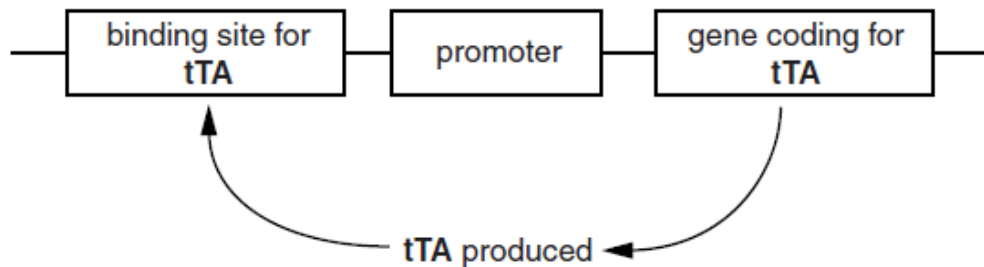


Fig. 3.1

- (i) Explain why, in gene technology, a promoter needs to be transferred along with the desired gene. [2]

- (ii) Switching on the gene coding for tTA in the mosquito larvae, rather than in the eggs, increases the effectiveness of this method of controlling mosquito numbers.

Explain why this is so. [2]

- (c) GM mosquitoes carrying the tTA gene can live and reproduce normally when fed on a diet containing an added chemical, **A**.

With reference to Fig. 3.1,

- (i) suggest how A could prevent death of the GM mosquitoes. [1]

- (ii) suggest why there is little danger of the gene carried by these GM mosquitoes being passed to other organisms from GM mosquitoes which escape or are released into the wild. [1]

In 1907, scientists found that a common plant tumour was caused by the invasion of a bacterium called *Agrobacterium tumefaciens*. They discovered that the tumour was caused when plasmids from the bacteria were taken into the DNA of the host plant's cells and expressed. By 1983, biotechnologists were able to introduce genes into plant cells using modified plasmids that do not result in tumour formation. Fig. 3.2 shows the procedure of transforming *Brassica* using this method.

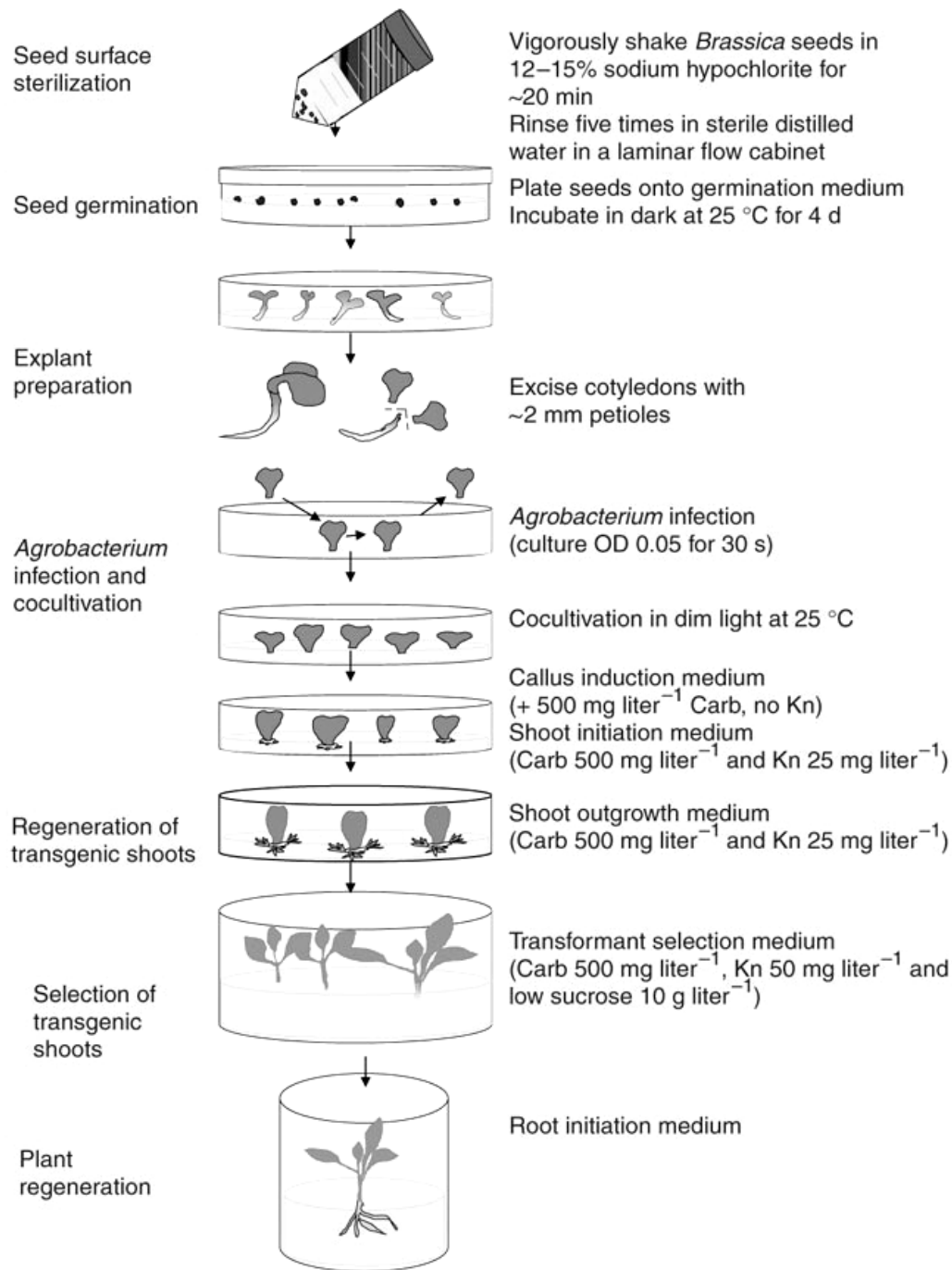


Fig. 3.2

- (d) With reference to Fig. 3.2, explain why carbenicillin and kanamycin were added in the various media. [3]

- (e) Explain why plant tissue culture technique is used to regenerate the transformed plants. [2]

- (f) State the difference in composition between the root initiation medium and the shoot outgrowth medium. [1]

- (g) Suggest why seeds were used instead of explants from adult *Brassica* plants, as starting plant material for this procedure. [1]

[Total: 14]

4 Planning question

The yield of many cultivated varieties of crop plants such as the onion, *Allium cepa*, is reduced by the presence of relatively high concentrations of salts, particularly sodium chloride, in the soil. There are, however, salt-tolerant varieties of onion that can be grown successfully in such places. One possible explanation for this is that the fluid in the cell vacuoles of plants of the tolerant varieties differs in solute potential from that of non-tolerant varieties.

Plan an investigation to find out whether or not the solute potentials of the cell vacuoles of epidermal cells of the bulbs of salt-tolerant onions differ from those of their non-tolerant counterparts.

You are provided with the following equipment. Choose your equipment from this list. You may **not** use any additional equipment.

- an unlimited supply of 1.0 mol dm^{-3} sodium chloride solution
- an unlimited supply of bulbs of two varieties of onion, one salt-tolerant and the other non-tolerant
- unlimited supply of distilled or deionised water
- beakers of various sizes
- glass specimen tubes
- microscope slides and coverslips
- volumetric flasks and measuring cylinders of various sizes
- graduated pipettes of various sizes, and pipette fillers
- dropping pipettes
- light microscope with low ($\times 100$), medium ($\times 200$) and high ($\times 400$) magnification and built-in illumination
- wash bottle
- marker pen
- stopwatch
- scalpel
- forceps
- glass rods for stirring
- mounted needles
- iodine in potassium iodide solution

Your plan should:

- include a clear statement of the hypothesis or prediction
- identify the key variables
- give full details and explanations of the procedures that you would adopt to ensure that the results are as precise and reliable as possible
- show how you would present and analyse your results
- include a brief risk assessment
- be written in clear scientific language, using the correct technical and scientific terms.

[Total: 12]

Free-response question

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 5**
- (a)** Outline how a named protein can be produced by genetic engineering for the treatment of diabetes mellitus. [6]
 - (b)** Discuss the benefits and associated ethical implications of the human genome project. [8]
 - (c)** Describe and give examples of unique features of zygotic stem cells, embryonic stem cells and adult stem cells. [6]